11. Although I haven't covered it explicitly in the text, note that every known particle has an *antiparticle*—a particle with the same mass but opposite force charges (like the opposite sign of electric charge). The electron's antiparticle is the positron; the up-quark's antiparticle is, not surprisingly, the anti-up-quark; and so on.

12. As we will see in Chapter 13, recent work in string theory has suggested that strings may be much larger than the Planck length, and this has a number of potentially critical implications—including the possibility of making the theory experimentally testable.

13. The existence of atoms was initially argued through indirect means (as an explanation of the particular ratios in which various chemical substances would combine, and later, through Brownian motion); the existence of the first black holes was confirmed (to many physicists' satisfaction) by seeing their effect on gas that falls toward them from nearby stars, instead of "seeing" them directly.

14. Since even a placidly vibrating string has *some* amount of energy, you might wonder how it's possible for a string vibrational pattern to yield a massless particle. The answer, once again, has to do with quantum uncertainty. No matter how placid a string is, quantum uncertainty implies that it has a minimal amount of jitter and jiggle. And, through the weirdness of quantum mechanics, these uncertainty-induced jitters have *negative*energy. When this is combined with the positive energy from the most gentle of ordinary string vibrations, the total mass/energy is zero.

15. For the mathematically inclined reader, the more precise statement is that the *square* of the masses of string vibrational modes are given by integer multiples of the square of the Planck mass. Even more precisely (and of relevance to recent developments covered in Chapter 13), the square of these masses are integer multiples of the *string scale* (which is proportional to the inverse square of the string length). In conventional formulations of string theory, the string scale and the Planck mass are close, which is why I've simplified the main text and only introduced the Planck mass. However, in Chapter 13 we will consider situations in which the string scale can be different from the Planck mass.